Shishir Shrestha

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EDUCATION

Rhodes College — Memphis, TN
Bachelor of Science — December 2025
Major in Computer Science; Minor in Mathematics, Major GPA: 3.95/4.0; Overall: 3.88/4.0
Relevant Coursework: Advanced Algorithms, Advanced Data Analytics, Linear Algebra, Artificial Intelligence, Computer Systems, Agent Based Modeling, Parallel Systems

Technical Skills

Programming Languages: C, C++, Python, Java, Haskell, SML, Javascript, Go, C#, R
Other: Tensorflow, Pytorch, Docker, Kubernetes, Git, Linux, Postgresql, Godot, Unity

Research Experience

Undergraduate Researcher
Rhodes College, Advisor: Prof. Sean Kugele, Department of Computer Science — **2023 - Present**
Mental Simulation in Agent Based Software Systems, Research Fellowship
- Surveyed and analyzed existing research literature on aphantasia and mental imagery, and contributed to extending the LIDA cognitive architecture to incorporate mechanisms accounting for these empirical findings.

Computational Cognitive Science, Summer Research Fellowship
- Developed a sensory memory module for the LIDA cognitive architecture using custom β-VAEs and vector databases
- Implemented a polyomino-based testing environment in Godot; and packaged the environment for public release to support reproducible research and external testing.
- Trained reinforcement learning agents using PPO, DQN, and DreamerV3, and benchmarked their performance in the custom polyomino environment to establish baselines for comparison with the LIDA agent.

Associative Memory Manifolds
- Developed a novel geometric framework for memory representation, modeling storage as deformations in a continuous n-dimensional manifold.
- Integrated perceptual encoding with attractor dynamics to simulate recall, interference, and depth-dependent forgetting consistent with the Ebbinghaus curve.
- Benchmarked against Sparse Distributed Memory, RNNs, and continuous attractor models, demonstrating alignment with cognitive phenomena such as graded interference, spacing effects, and false recall.
- Investigating methods for developing neural embedding systems that can acquire new knowledge while preserving the geometric consistency of previously learned representations.
- Addressing the challenge of catastrophic drift in encoder spaces by exploring mechanisms for stability, plasticity, and retention of semantic structure.
- Designing approaches that ensure perceptual similarity and temporal proximity map coherently in embedding space, enabling robust downstream memory and reasoning tasks.

Undergraduate Researcher
Rhodes College, Advisor: Prof. Ibrahim Abdelrazeq, Department of Mathematics — **2025 - Present**
- Implemented and verified a Lévy-driven Ornstein–Uhlenbeck (CAR(1)) model, applying methodological tests to confirm model assumptions and driving process properties, and tested the framework on dynamic stock spread data from the S&P 500
- Developing and implementing the Continuous-Time ARMA CARMA(2,1) model for empirical evaluation, including estimation under Lévy-driven dynamics and simulation-based testing

Undergraduate Researcher
Rhodes College, Advisor: Prof. Betsy Sanders, Department of Computer Science — **2022 - 2023**
- Designed a one-to-one VR recreation of the test environment in Unity Game Engine
- Implemented various performance metrics to track and measure user’s actions in the VR and real environment

- Validated established depth perception models by analyzing user interaction data, replicating key findings from prior literature.

Papers

- Shrestha, S., & Kugele, S. (Under Review). Memory Manifolds: A Neurobiologically Inspired Model. Proceedings of the 48th Annual Conference of the Cognitive Science Society. Cognitive Science Society.

Poster Presentation

A Human Newton's Cradle

- Designed and presented a large-scale physics demonstration replicating Newton's cradle using human participants in bubble suits.
- Developed computer simulations to model momentum transfer and optimize experimental setup prior to the live demonstration. The project culminated in a public presentation and promotional video showcasing both the physical experiment and its educational impact.

Selected Projects

Optimized N-Body Simulation with OpenSHMEM [C, OpenSHMEM]

- Implemented a distributed memory parallel algorithm to efficiently compute gravitational interactions.
- Leveraged OpenSHMEM for inter-process communication, reducing synchronization overhead and improving scalability.
- Optimized data structures and memory access patterns to enhance performance on multi-node systems.

Interpreter Development in Standard ML (SML**)**

- Developed a recursive descent parser to handle language syntax and structure.
- Utilized functional programming paradigms to create a concise and maintainable interpreter.

Texas Hold'em Poker Solver (Go**)**

- Developed a poker solver for No-Limit Texas Hold'em in Go implementing hand evaluation, game tree traversal, and equilibrium computation.
- Optimized for concurrency and memory efficiency using goroutines and channel-based parallelism.
- Integrated REST-style API support for programmatic interaction and external analysis tools

Picobytes - Execute and Test Service [Python, C, Docker, Kubernetes, FastAPI, Bash]

- Designed a robust API to provide execution results, including compiler errors with line numbers and test case outcomes.
- Engineered safeguards to ensure secure and efficient code execution, improving system reliability and performance.
- Leveraged Docker and Kubernetes for containerized, sandboxed environments, ensuring secure and isolated code execution.

Work Experience

Rhodes Student Associate - A competitive student employment program offering professional experience within campus departments

Information Technology and Library Services, Paul Barret Jr. Library, Rhodes College **2023 - Present**

- Managed Help Desk operations, including scheduling, training, and technical support for a 20–30 member student team.
- Provided IT and library support, resolving service requests, maintaining documentation, and assisting with campus technology troubleshooting.

Other Activities

- International Collegiate Programming Contest (2024,2025), Advent of Code 2020-2025, and various hackathons
- International Mathematical Olympiad, National Finalist
- Game Jam 2024, Best Game Mechanic